Q29 Ans: C

Most particles passed through undeflected or with a small deflection

Since very few of the particles were scattered through large angles, the probability of the particle getting close to the centre of the positive charge is small. This shows that the atom consists of mostly empty space.

Small fraction (<1%) of α-particles are deflected through large angles;

To produce such a large deflection, there must be a large force

All the positive charges in the atom are concentrated as a nucleus in a small region of space as compared to the diameter of the atom.

Few particles are reflected backwards, through an angle close to 180°

As such, the nucleus is small and very massive.

Q30 Ans: D